In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import
seasonal_decompose

# Create sample website traffic data
np.random.seed(42)

dates =
pd.date_range(start='2025-01-01',
periods=120, freq='D') page_views =
(
 500
 + np.arange(120) * 2
 + 50 * np.sin(2 * np.pi *
np.arange(120) / 7)  +
np.random.normal(0, 20, 120)
)

# Add a few traffic spikes
page_views[40] += 200
page_views[85] += 250

data = pd.DataFrame({
 'Date': dates,
 'Page_Views': page_views
})

data.set_index('Date', inplace=True)

# Display the dataset
print("Website Traffic Data:")
print(data.head())

# 1. Time Series Decomposition
decomposition = seasonal_decompose(
 data['Page_Views'],
 model='additive',
 period=7
)

decomposition.plot()
plt.suptitle("Time Series
Decomposition", y=1.02)
plt.tight_layout()
plt.show()

# 2. Moving Average
data['Page_Views_MA'] =
data['Page_Views'].rolling(
window=7
).mean()

data[['Page_Views',
'Page_Views_MA']].plot(
 figsize=(10, 5)
)
plt.title("Page Views with Moving
Average")
plt.xlabel("Date")
plt.ylabel("Page Views")
        plt.show()

        # 3. Anomaly Detection using Z-score
        mean = data['Page_Views'].mean()
        std = data['Page_Views'].std()

        data['Page_Views_Z'] = (
         data['Page_Views'] - mean
        ) / std

        anomalies = data[data['Page_Views_Z'].abs() > 3]

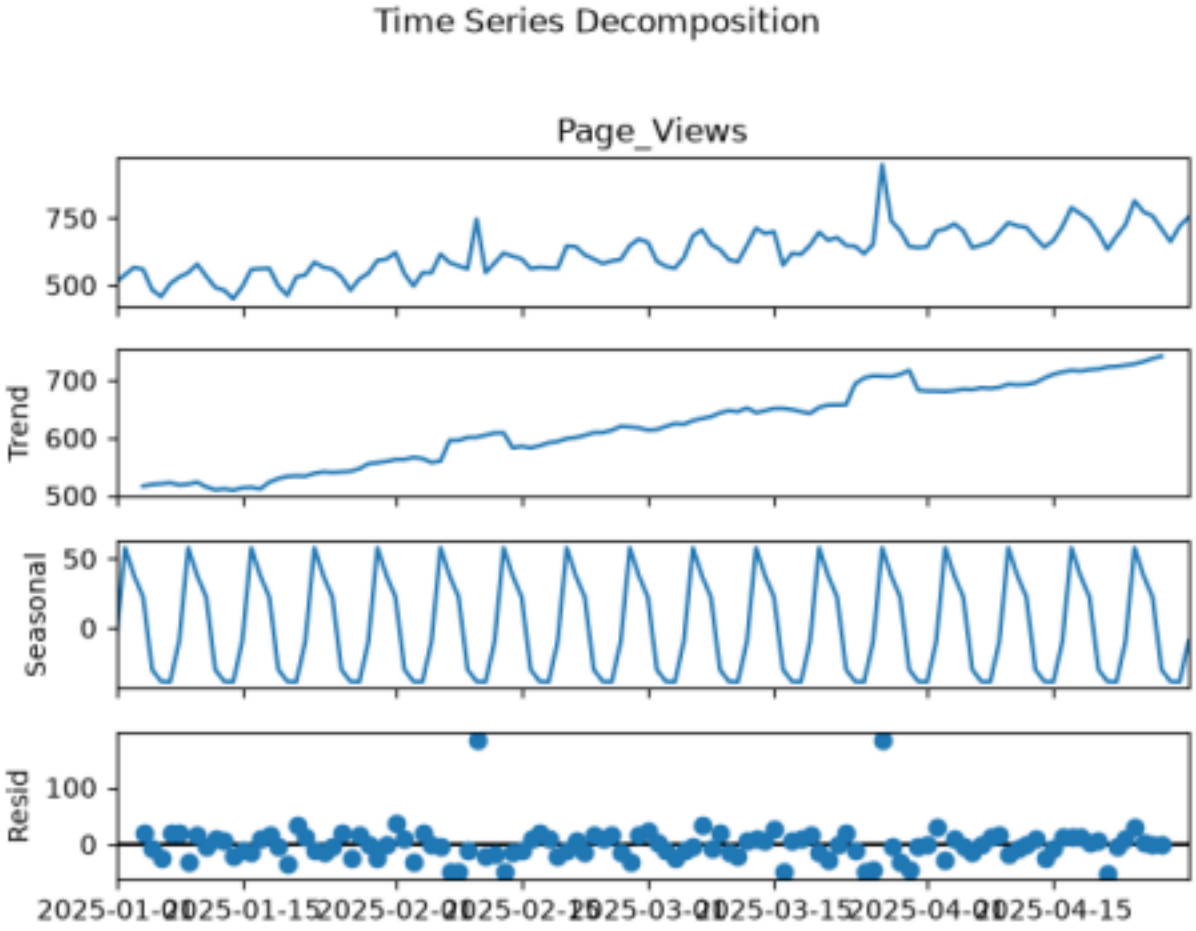
        print("\nDetected Anomalies:")
        print(anomalies[['Page_Views', 'Page_Views_Z']])

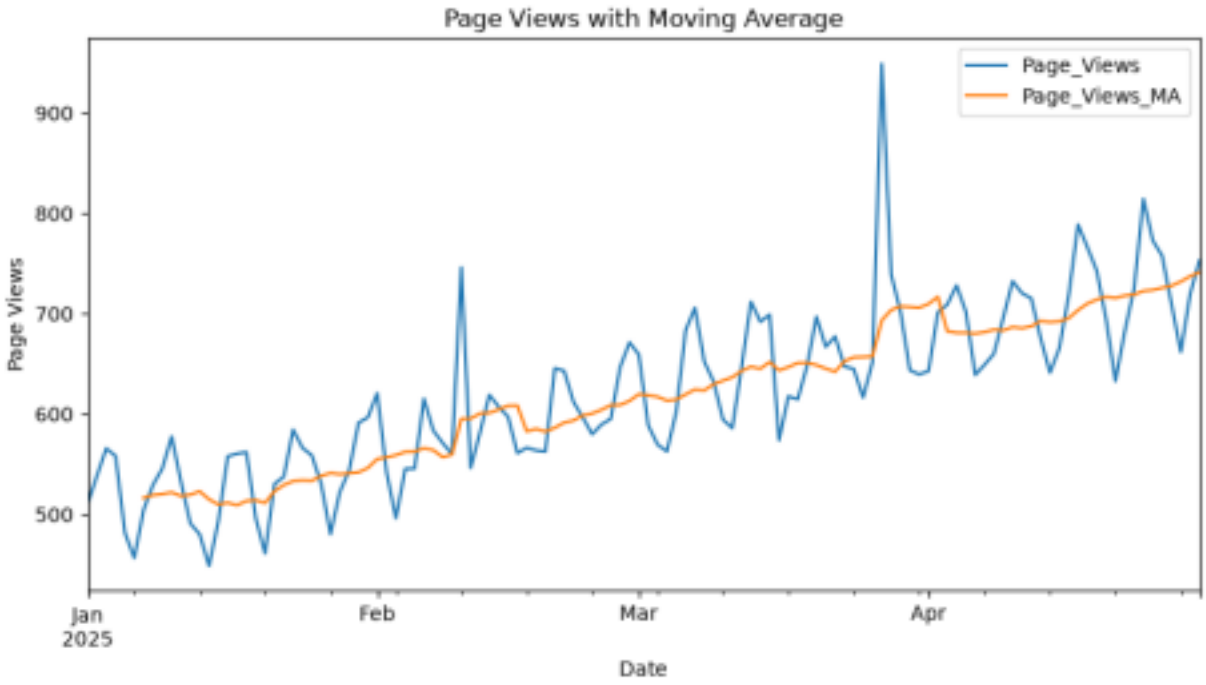
        # Plot anomalies
        plt.figure(figsize=(10, 5))
        plt.plot(
         data.index,
         data['Page_Views'],
         label='Page Views'
        )
```

```
 plt.scatter(
  anomalies.index,
  anomalies['Page_Views'],
  color='red',
  label='Anomalies'
 )

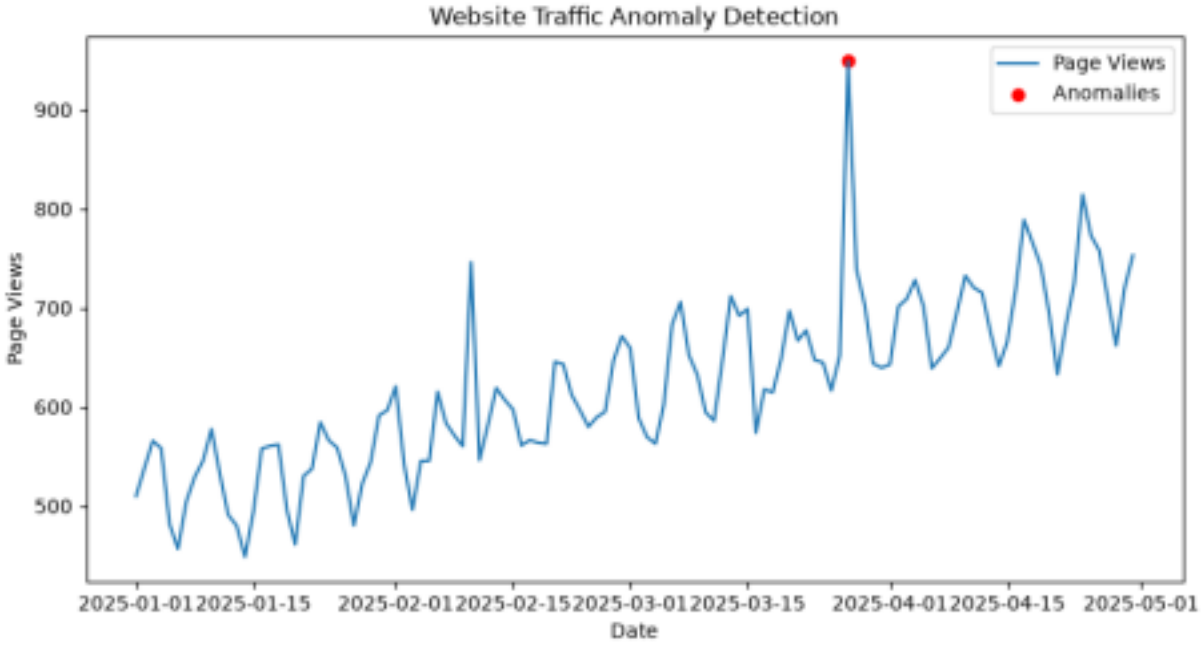
 plt.title("Website Traffic Anomaly Detection")
 plt.xlabel("Date")
 plt.ylabel("Page Views")
 plt.legend()
 plt.show()
```

```
Website Traffic Data:
 Page_Views
Date
2025-01-01 509.934283
2025-01-02 538.326288
2025-01-03 565.700166
2025-01-04 558.154784
2025-01-05 481.622746
```






```
Detected Anomalies:
  Page_Views Page_Views_Z
 Date
 2025-03-27 949.056433 3.823627
```



In [ ]: